

PES 2020 / 2021 — Overall Rating Formula (draft)

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1. Top-level pipeline

The overall rating is computed for a player *at a query position*. By default the query position is the player's registeredPosition. The pipeline branches on whether the player is being rated at his registered position or at another position.

Overall = **clamp(40, ∞)(finalRating)**, where:

```
if (queryPosition === registeredPosition):
    base = weightedAverage(player, queryPosition) + positionBonus[queryPosition]
    if queryPosition == 'GK':
        finalRating = base
    else:
        curved = curvedSumAverage(player) // see §4
        finalRating = floor( (base*60 + curved*40 + 50) / 100 )
else:
    // rated out of position
    base = weightedAverage(player, queryPosition)
    base += offPositionBonus[queryPosition][positionRating] // 'A' / 'B' / 'C'
    finalRating = base // no curved blend
```

All arithmetic uses BigInt truncation (integer division), so divisions effectively round *toward zero* — but the +50 offset before /100 turns it into nearest-integer rounding for positive sums.

2. Weighted average

The weighted average is the position-specific sum of (*ability* – *baseline*) × *weight*, divided by 100:

$$\text{weightedAverage}(\text{player}, \text{position}) = \text{floor}\left(\left(\sum_a (S_a - \text{offset}_a) \times W_{\text{position},a} + 50 \right) / 100 \right)$$

- S_a — the player's value of ability a (40–99 for normal abilities, 1–4 for Weak Foot Accuracy).
- offset_a — 1 if a = Weak Foot Accuracy, else 25.
- $W_{\text{position},a}$ — the position-specific weight (see §5).

2.1 Composite GK abilities

Five GK abilities are not summed individually. They are folded into two composites and added with the position weights of GKAwareness and GKReflexes:

$$\begin{aligned} \text{GKAwa}_{\text{composite}} &= \lfloor (\text{GKAwareness} + \text{GKReach}) / 2 \rfloor \\ \text{GKRef}_{\text{composite}} &= \lfloor (\text{GK-catching} + \text{GK-clearing} + \text{GKReflexes}) / 3 \rfloor \\ \text{contribution} &= (\text{GKAwa}_{\text{composite}} - 25) \times W_{\text{GKAwareness}} + (\text{GKRef}_{\text{composite}} - 25) \times W_{\text{GKReflexes}} \end{aligned}$$

3. Sum average (used only for the curve, non-GK positions only)

The sum average is a flat mean across 22 contributors: 20 individual outfield abilities + 2 GK composites. Weak Foot Accuracy is excluded.

$$\text{sumAverage}(\text{player}) = \lfloor (\lfloor \text{sum} \times 100 / 22 \rfloor + 50) / 100 \rfloor$$

where *sum* is the 22-contributor total described below.

The 20 outfield abilities (each contributes raw value):

OffensiveAwareness, BallControl, Dribbling, TightPossession, LowPass, LoftedPass, Finishing, Heading, PlaceKicking, Curl, Speed, Acceleration, KickingPower, Jump, PhysicalContact, Balance, Stamina, DefensiveAwareness, BallWinning, Aggression

Plus the 2 GK composites (same as §2.1): $\lfloor (\text{GKAwa} + \text{GKReach}) / 2 \rfloor$ and $\lfloor (\text{GKCat} + \text{GKClr} + \text{GKRefl}) / 3 \rfloor$.

4. Curve mapping $\text{mapRating}(c)$

The sum average is fed through a non-linear curve before being blended with the weighted average:

Sum avg <i>c</i>	Mapped value	Δ
$c \leq 61$	$c - 9$	-9
$c = 62$	$c - 6$	-6
$c = 63$	$c - 4$	-4
$c = 64$	$c - 2$	-2
$c = 65$	c	0
$c = 66$	$c + 1$	+1
$c = 67$	$c + 3$	+3
$c = 68$	$c + 5$	+5
$c = 69$	$c + 6$	+6
$c = 70$	$c + 7$	+7
$c = 71$	$c + 8$	+8
$72 \leq c \leq 73$	$c + 10$	+10
$74 \leq c \leq 75$	$c + 12$	+12
$76 \leq c \leq 77$	$c + 13$	+13
$c \geq 78$	$c + 14$	+14

The curve compresses ratings near 60 (penalty up to -9) and lifts ratings near 70+ (bonus up to +14). It only affects non-GK final ratings.

5. Position-specific ability weights

Wide rows; columns aggregated across positions. 0 values shown grey for readability.

Ability \ Position	GK	CB	LB/RB	DMF	CMF	LMF/RMF	AMF	LWF/RWF	SS	CF
Offensive Awareness	0	0	6	7	5	7	14	17	15	31
Ball Control	0	0	9	18	23	15	23	19	19	23
Dribbling	0	0	9	10	16	17	17	15	13	9
Tight Possession	0	0	5	5	8	8	7	7	6	5
Low Pass	0	0	0	18	23	7	21	5	9	0
Lofted Pass	0	0	14	19	21	12	14	9	9	0
Finishing	0	0	0	0	0	0	17	11	14	36
Heading	0	22	0	0	0	0	0	0	0	3
Place Kicking	0	0	0	0	0	0	0	0	0	0
Curl	0	0	0	12	0	4	0	0	0	0
Speed	0	10	16	3	4	24	5	15	9	5
Acceleration	0	0	14	3	6	21	7	15	21	5
Kicking Power	0	0	0	0	0	0	0	5	6	3
Jump	11	19	11	5	0	0	0	0	0	3
Physical Contact	11	19	8	9	3	0	2	2	2	8
Balance	0	0	4	4	2	0	3	4	5	2
Stamina	0	9	14	14	17	13	3	6	4	0
Defensive Awareness	0	25	14	7	3	0	0	0	0	0
Ball Winning	0	17	8	3	0	0	0	0	0	0
Aggression	0	8	5	1	0	0	0	0	0	0
GK Awareness ($\times$ [(GKAwa+GKReach)/2])	50	0	0	0	0	0	0	0	0	0
GK Reflexes ($\times$ [(GKCat+GKClr+GKRefl)/3])	48	0	0	0	0	0	0	0	0	0
Weak Foot Accuracy (offset = 1)	0	23	56	23	23	56	37	47	37	38
Sum of weights	120	152	193	161	154	184	170	177	169	171

Note that abilities not in the table (*Weak Foot Usage, Form, Injury Resistance*) do **not** contribute to the rating in this implementation.

6. Position bonuses

6.1 At registered position

Position	Bonus
GK	+8
CB	+8
LB	+9
RB	+9
DMF	+9
CMF	+8
LMF	+8
RMF	+8
AMF	+8
LWF	+10
RWF	+10
SS	+9
CF	+8

Applied *before* blending with the curved sum average.

6.2 Off-position bonuses (rating A / B / C)

Position	A	B	C
GK	+5	+3	0
CB	+5	+3	0
LB	+4	+2	0
RB	+4	+2	0
DMF	+4	+2	0
CMF	+5	+3	0
LMF	+5	+3	0
RMF	+5	+3	0
AMF	+5	+3	0
LWF	+4	+2	0
RWF	+4	+2	0
SS	+5	+3	0
CF	+5	+3	0

Off-position rating **does not** apply the curved-sum blend; it just returns the bonus-adjusted weighted average (clamped to ≥ 40).

7. Branching summary

Case	Final formula
GK at registered position	$\max(40, \text{weightedAvg} + 8)$
Outfield at registered position	$\max(40, \lfloor ((\text{weightedAvg} + \text{posBonus}) \cdot 60 + \text{curvedSumAvg} \cdot 40 + 50) / 100 \rfloor)$
Any role rated off-position	$\max(40, \text{weightedAvg} + \text{offPositionBonus}[A B C])$

8. Worked example — outfield CMF

Hypothetical CMF with stats: BallControl 85, LowPass 84, LoftedPass 80, Dribbling 78, Stamina 82, OffAwa 75, TightPos 72, Speed 70, Acceleration 73, PhysCon 68, Balance 70, DefAwa 65, WeakFootAcc 3, plus zeros where the CMF weight is 0.

Weighted contribution (only non-zero weights matter for CMF):

```

(75-25)·5 + (85-25)·23 + (78-25)·16 + (72-25)·8 + (84-25)·23 + (80-25)·21
+ (70-25)·4 + (73-25)·6 + (68-25)·3 + (70-25)·2 + (82-25)·17 + (65-25)·3
+ (3-1)·23
= 250 + 1380 + 848 + 376 + 1357 + 1155
+ 180 + 288 + 129 + 90 + 969 + 120
+ 46
= 7188

```

```

weightedAvg = [(7188 + 50) / 100] = 72
+ position bonus (CMF) = 72 + 8 = 80

```

```

sumAverage ≈ [(sum·100/22) + 50]/100 // sum of 22 contributors
            = (let's say) 73
curvedSumAvg = mapRating(73) = 73 + 10 = 83

```

```

final = [(80·60 + 83·40 + 50) / 100]
       = [(4800 + 3320 + 50) / 100]
       = [8170 / 100]
       = 81

```

9. Quirks & caveats

- **Integer truncation.** All divisions use BigInt truncation. Adding +50 before /100 simulates round-half-up only because the dividend is positive.
- **Goalkeepers skip the curve.** The 60/40 blend is only applied for non-GK at the registered position.
- **No curve when off-position.** Off-position ratings just take the weighted average plus the small A/B bonus, so a winger rated as a CB will look much weaker than an actual CB with similar stats.
- **Excluded abilities.** Weak Foot *Usage*, Form, and Injury Resistance do not enter the formula.
- **WFA offset.** Weak Foot Accuracy is on a 1–4 scale, so its baseline is 1 instead of 25; this is hard-coded as a special case in `getWeightedAverage`.
- **SB shared, SMF shared, WF shared.** LB/RB share one weight table (SB); LMF/RMF share SMF; LWF/RWF share WF.
- **Floor of 40.** Final result is `max(rating, 40)` — the rating cannot drop below 40 even for very poor stats.